

MUHAMMAD TAHA CHEEMA

+1 (548) 390-4499 | [✉ tahacheema95@gmail.com](mailto:tahacheema95@gmail.com) | [🌐 linkedin.com/in/tahacheema](https://www.linkedin.com/in/tahacheema) | [🐙 github.com/tahachm](https://github.com/tahachm)

EDUCATION

University of Waterloo

Waterloo, Canada

Master of Mathematics in Computer Science (Thesis) **Grade: 99.25/100**

Jan 2026 – Dec 2027 (expected)

- Coursework: Adv. Topics in SE - Large Language Models for SE, Adv. Topics in SE - SE for Big Data and AI.

Lahore University of Management Sciences (LUMS)

Lahore, Pakistan

Bachelor of Science in Computer Science **Grade: 3.97/4.0**

Sep 2021 – May 2025

- Coursework: Distributed Systems, Data Structures, Algorithms, Databases, Adv. Topics in Internet Research.
- Awards: Graduated with High Distinction, Dean's Honor List (2021–2025), LUMS Merit Scholarship (2021–2025).

Stanford University

California, USA

Exchange Quarter **Grade: 4.0/4.3**

Jun 2023 – Aug 2023

- Coursework: CS 229 - Machine Learning, ENGR 145 - Technology Entrepreneurship.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL, Golang

Web & UI: React, Next.js, Redux, Recoil, Material UI, tRPC, Motion

Backend & APIs: REST APIs, Node.js, PostgreSQL, MongoDB, Prisma ORM, Supabase

Cloud & Tooling: AWS, Vercel, Railway, Git

AI/ML: LLM pipelines & orchestration, vector & semantic search, RAG, Meilisearch, Milvus, Hugging Face

EXPERIENCE

University of Waterloo

Waterloo, Canada (on-site)

Graduate Student Researcher

Jan 2026 – Present

- Architected an **over-the-air testbed** for non-terrestrial (NTN) 5G within a **cross-functional CS/ECE team**, using **agentic coding tools** to port custom channel-impairment models into **OpenAirInterface's** simulator in **C**.
- Ran experiments in the simulator to **scope 6 actionable NTN optimization avenues** for the team to work on, utilizing **Linux** networking commands such as **ip netns** to isolate network namespaces and **tc netem** to inject packet delay.

GreatFrontEnd

Singapore (remote)

Software Engineer

Aug 2025 – Jan 2026

- **Led AI engineering** on their new product, Socialmon, an AI companion for brands to create viral marketing ideas. Programmed and **deployed a scalable, real-time Next.js artifact search engine** on Railway, combining **vector and lexical search** from Meilisearch's JavaScript SDK with a custom **re-rank algorithm** to ensure semantic variation among top search hits. Scaled to **118k data points with search latencies under 50ms** on a test dataset.
- Built a pipeline to generate ad and social-media post layouts from natural-language prompts, using **LLM function calling** to enforce structured JSON output. **Extracted high-fidelity JSON representations** of public viral posts via background masking and **Gemini's image segmentation**, then used them as **few-shot examples** to hold aesthetic quality while shipping **fully editable HTML designs** rather than the flat, uneditable images that image models produce.

GreatFrontEnd

Singapore (remote)

Software Engineering Intern

Feb 2025 – Apr 2025

- Built an in-browser **executable JS IDE** into their **Node.js** interview-preparation platform by **running code in a sandboxed iframe**, enabling learners to practice coding without navigating away from the page and **increasing time-on-page by 1.4x**.

AI Centre for Educational Technologies

Singapore (on-site)

Software Engineering Intern

May 2024 – Aug 2024

- Engineered SoftMark, a **productionized digital grading platform** deployed at **NUS** and secondary schools in Singapore (**TypeScript, React, Golang, PostgreSQL**), including a **2D grid allocation component** I designed and shipped **end-to-end** that **reduced time spent on grading workload distribution by 30%**.
- **Devised a trie-based algorithm over open-source OCR** to detect question numbering on answer sheets, **doubling teachers' productivity** by filtering unattempted questions.

Distributed and AI Systems Lab, LUMS

Lahore, Pakistan (on-site)

Research Assistant

Sep 2024 – Apr 2025

- Designed and benchmarked **TweakLLM (arXiv:2507.23674)**, an **adaptive caching architecture for LLM systems** that uses a lightweight model to dynamically adapt cached responses to similar user queries, **cutting compute costs by up to 68%** on real-world user-LLM interaction datasets.
- Used an **LLMs-as-jury evaluation harness** of three **GPT-4o judge personas** to rank **1,000 response pairs**.

PROJECTS

ForeSite: An Agentic Residential Community Simulation for Neighbourhood Design

Jun 2026

Third Place, University of Waterloo Future Cities Innovation Challenge

University of Waterloo

- Built and pitched a neighbourhood simulation using **LLM personas**, extending an open-source LLM-agent engine (**TypeScript, React**), winning **third place out of 40 teams** in a two-week competition. Used a **Python** analytics pipeline and a **controlled A/B experiment** to show **3x more daily social isolation** in a sprawl neighbourhood layout than in a walkable one.

Digitizing Partition-Era Abduction Registers

Oct 2025 – Dec 2025

Research Contract - School of Humanities and Social Sciences

LUMS

- Orchestrated Google's **Gemini API** to **digitize 37,210 entries** from 1940s India-Pakistan partition abduction registers, turning the historical archive into an **indexable, searchable** dataset for learning and research.

ACHIEVEMENTS AND HONORS

2026 – 2027: International Master's Award of Excellence valued at C\$12,500

University of Waterloo

2023: 70% Tuition Waiver Award

Stanford University